QUIZ

NumPy & Pandas

Python Data Science Series

STUDENT INFORMATION

FULL NAME

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CLASS / SECTION

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ROLL NUMBER

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Date: _____23/2/2026_____

Total Marks: ____ / 30

SECTION 1

NumPy Arrays & Operations

Questions 1–5 · Broadcasting, Indexing & Numerical Computing

5 Questions × 1 Mark = 5 Marks

1

ARRAY OPERATIONS

[1 Mark]

What does the following NumPy operation return?
np.array([1, 2, 3]) * 3

A	[3, 6, 9]	**B**	[1, 2, 3, 1, 2, 3, 1, 2, 3]
C	[3, 3, 3]	**D**	Error – cannot multiply array by scalar

Answer: _____A_

2

BROADCASTING

[1 Mark]

Which of the following NumPy array shapes are broadcast-compatible?

A	(3, 4) and (5, 4)
B	(3,) and (3, 3)
C	(4, 1) and (1, 4)
D	Both B and C

Answer: _____B_

3

INDEXING & SLICING

[1 Mark]

Given arr = np.array([[1,2,3],[4,5,6],[7,8,9]]), what does arr[1:, ::2] return?

A [[4, 6], [7, 9]]

B [[4, 5, 6], [7, 8, 9]]

C [[4, 6], [7, 9]]

D [[5, 6], [8, 9]]

Answer: _____A_

4

NUMERICAL COMPUTATIONS

[1 Mark]

Which NumPy function computes the dot product of two 1-D arrays?

A np.multiply()

B np.dot()

C np.cross()

D np.sum()

Answer: _____B_

5

ARRAY OPERATIONS

[1 Mark]

What is the output of np.arange(2, 10, 3)?

A [2, 5, 8, 11]

B [2, 5, 8]

C [3, 6, 9]

D [2, 4, 6, 8]

Answer: _____B_

SECTION 2

Pandas — Series & DataFrames

Questions 6–12 · Data Loading, Filtering, Sorting & Wrangling

7 Questions × 1 Mark = 7 Marks + 3 Short Answer × 2 Marks = 6 Marks

Which Pandas method is used to read a CSV file into a DataFrame?

A	pd.read_table()
B	pd.read_csv()
C	pd.load_csv()
D	pd.import_csv()

Answer: _____B_

7

DATA LOADING

[1 Mark]

Which parameter in pd.read_excel() specifies which sheet to load?

A tab_name

B sheet_index

C sheet_name

D worksheet

Answer: _____C_

8

FILTERING

[1 Mark]

Given df with a column 'Age', which expression filters rows where Age > 25?

A df[df.Age > 25]

B df.filter(Age > 25)

C df.where('Age > 25')

D df.query['Age > 25']

Answer: _____A_

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SORTING & GROUPING

[1 Mark]

Which method sorts a DataFrame by the 'Score' column in descending order?

A df.sort('Score', desc=True)

B df.order_by('Score', ascending=False)

C df.sort_values('Score', ascending=False)

D df.arrange('Score', reverse=True)

Answer: _____C_

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GROUPING

[1 Mark]

What does df.groupby('Dept')['Salary'].mean() return?

A A single mean salary value

B Mean salary grouped by department

C Mean salary of the 'Dept' column

D Error – cannot group then mean

Answer: _____B_

Which method combines two DataFrames horizontally (column-wise)?

A pd.concat([df1, df2], axis=0)

B pd.concat([df1, df2], axis=1)

C df1.append(df2)

D pd.merge(df1, df2, how='outer')

Answer: _____B_

What does df.dropna() do to a DataFrame?

A Drops all columns with NaN values

B Fills NaN values with 0

C Drops all rows containing NaN values

D Replaces NaN with the column mean

Answer: _____C_

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PANDAS – GROUPBY

[2 Marks]

SHORT ANSWER

Explain the split-apply-combine strategy used by groupby() in Pandas. Provide a real-world example showing how groupby() with an aggregation function is used.

Answer: The split-apply-combine strategy used by groupby() in Pandas works in three simple steps. First, the data is **split** into groups based on a column, such as grouping employees by department. Next, a function is **applied** to each group, like calculating the average or maximum score. Finally, the results are **combined** into a new summarized table. For example, if HR wants to find the average performance score per department, they can use df.groupby("Department")["Score"].mean(). This groups employees by department, calculates the average score for each group, and returns a clear summary that helps management understand departmental performance.

SHORT ANSWER

What is broadcasting in NumPy? State the broadcasting rules and give an example showing how arrays of shapes (3, 1) and (1, 4) would interact.

Answer: Broadcasting in NumPy allows arrays of different shapes to perform operations automatically. Rules: dimensions must match or one must be 1. Comparison starts from the right. For shapes (3,1) and (1,4), NumPy expands them to (3,4), repeating values, allowing element-wise operations like addition across rows and columns.

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DATA WRANGLING

[2 Marks]

SHORT ANSWER

Describe any TWO common data-wrangling tasks in Pandas. For each task, name the method used and write a one-line code example.

Answer:One common data-wrangling task is handling missing data using dropna(), for example: df = df.dropna(), which removes rows with null values. Another task is filtering data using conditions, such as df = df[df["Score"] > 80], which selects only rows where the score is greater than 80.

END OF QUIZ

Please review all answers before submitting.

Section	Topic	Questions	Marks
Section 1	NumPy Arrays & Operations	Q1–Q5	5 Marks
Section 2	Pandas MCQ	Q6–Q12	7 Marks
Section 2	Short Answer	Q13–Q15	6 Marks
	TOTAL		**30 Marks**